

Asymmetric synthesis

不对称合成

参考书: Organic Chemistry, p1102-1133, Clayden, Greeves, Warren and Wothers, Oxford University Press, 2001

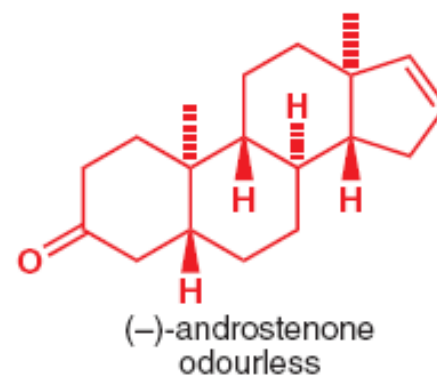
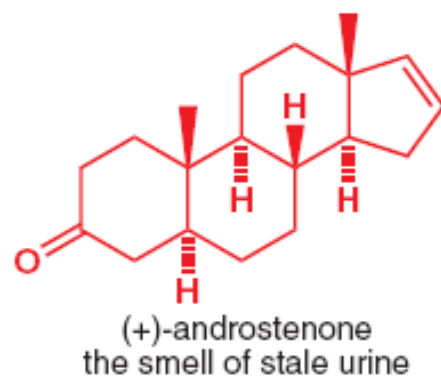
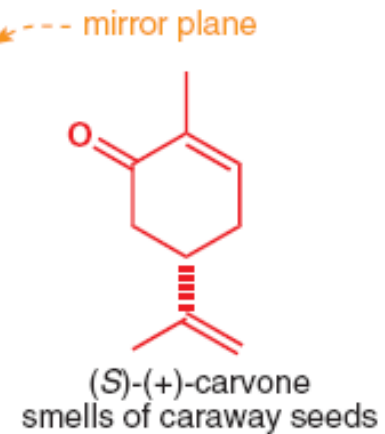
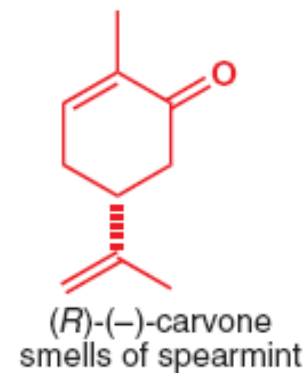
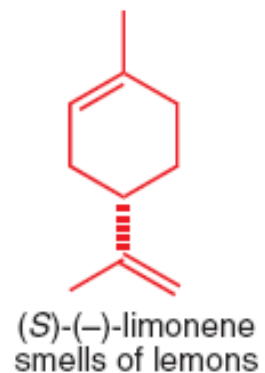
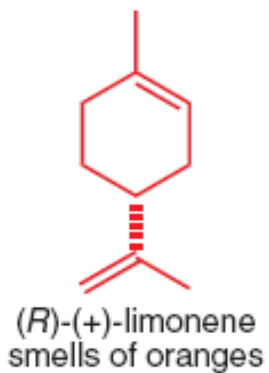




天然产物对映体

enantiomeric smells

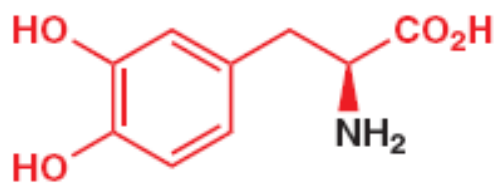
mirror plane



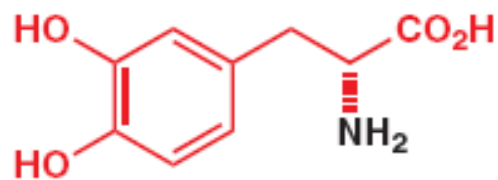
mirror plane is
plane of paper



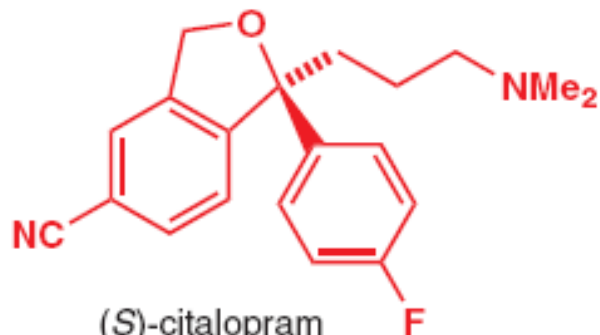
手性药物分子



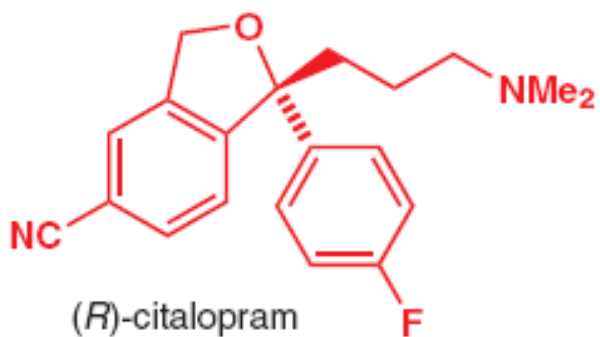
L-dopa
treatment for
Parkinson's disease



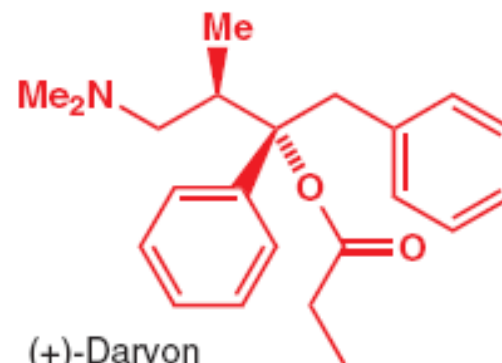
D-dopa
is toxic



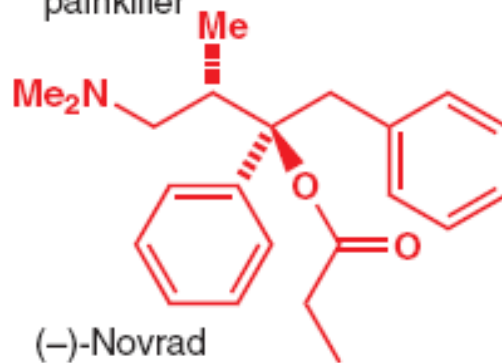
(*S*)-citalopram
drug for depression



(*R*)-citalopram
<1% antidepressant effect of (*S*) enantiomer



(+)-Darvon
painkiller



(-)-Novrad
cough suppressant



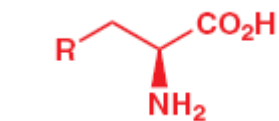
不对称合成主要策略

- 手性池
- 手性拆分
- 手性辅基
- 手性配体-金属复合物不对称催化
- 手性有机小分子不对称催化

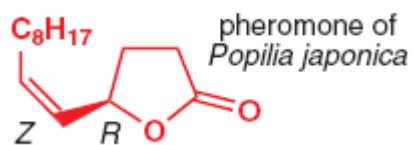


手性池策略

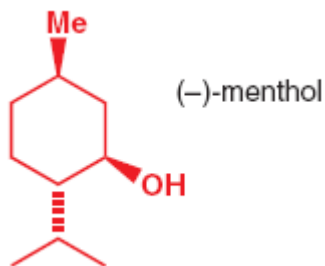
自然界存在的手性分子



氨基酸



外激素



薄荷醇

手性池

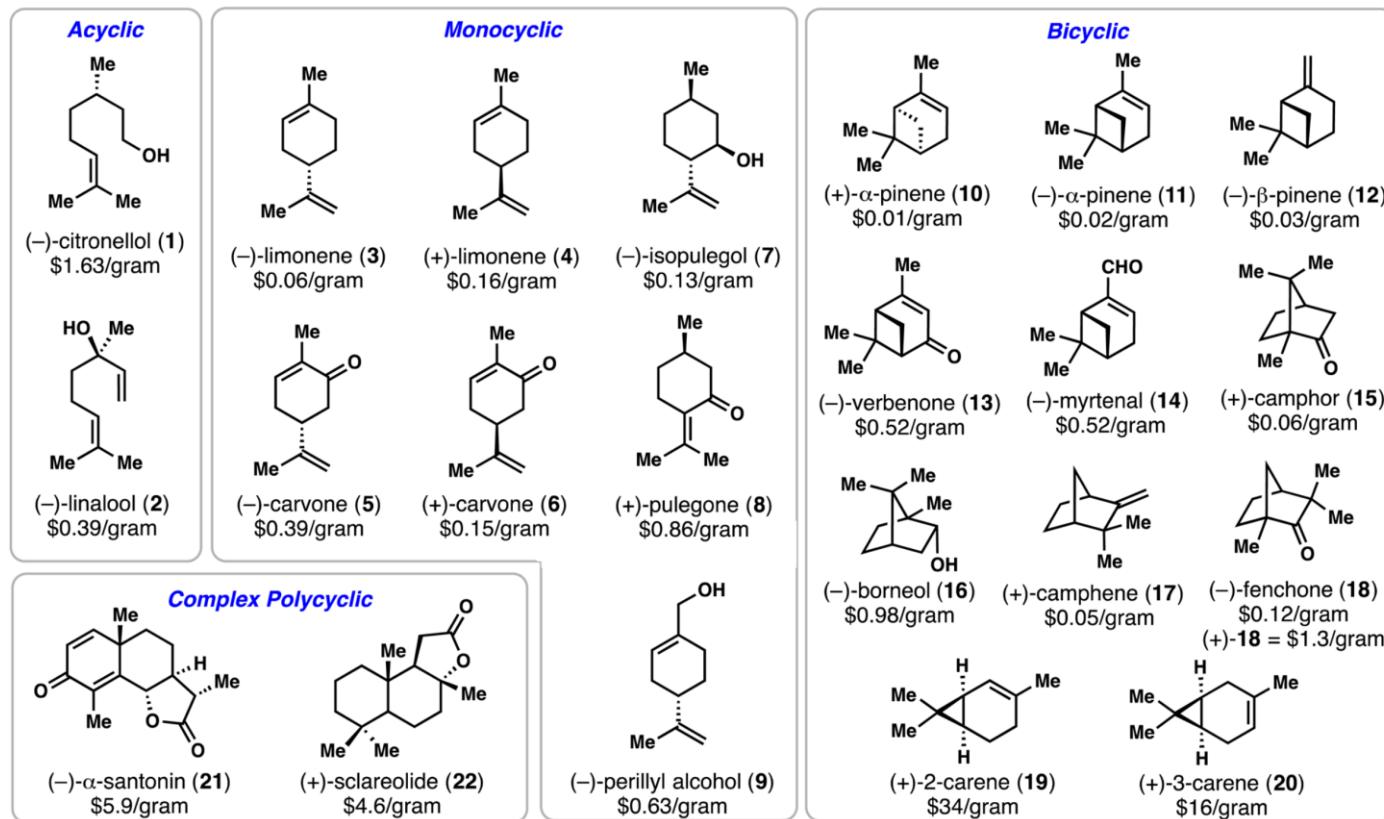
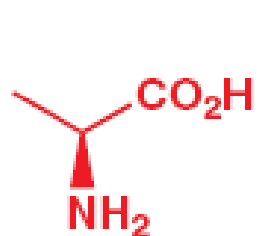


Figure 1. Chiral pool terpenes of both historical use and modern use in natural product synthesis.

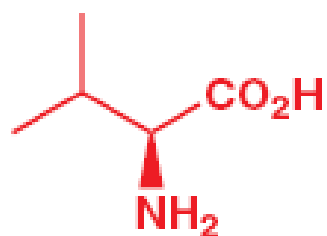


手性池策略

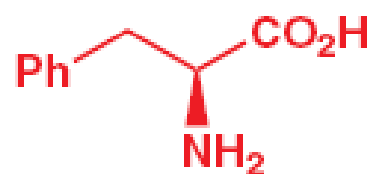
天然氨基酸



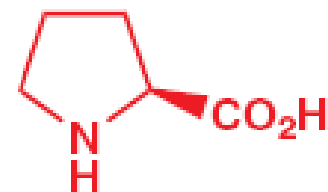
(S)-(+)-alanine



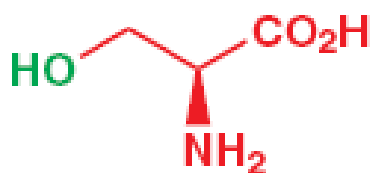
(S)-(+)-valine



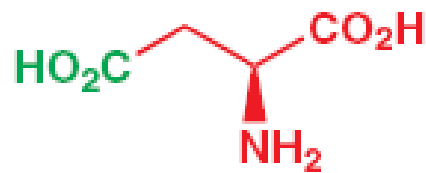
(S)-(-)-phenylalanine



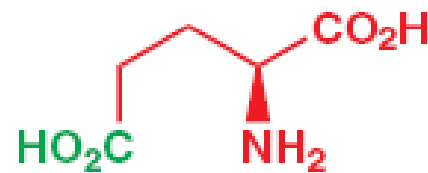
(S)-(-)-proline



(S)-(+)-serine



(S)-(+)-aspartic acid

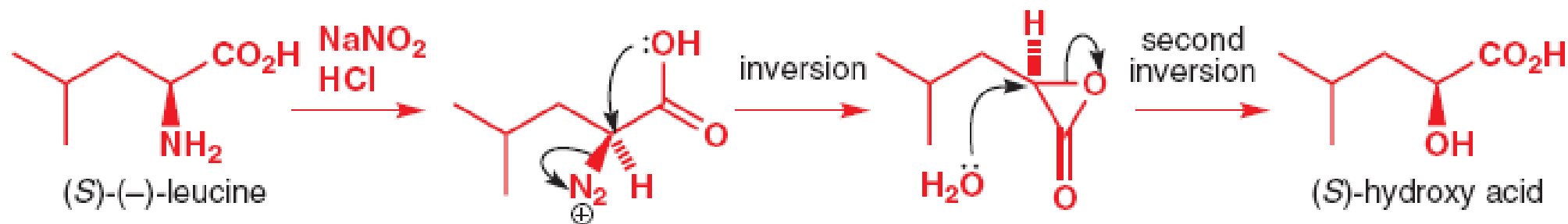


(S)-(+)-glutamic acid



从手性氨基到手性羟基

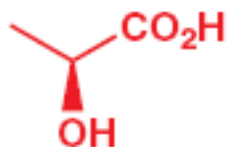
diazotization–hydrolysis of amino acids to give hydroxy acids proceeds with overall retention



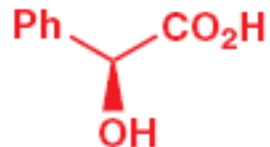


从手性氨基到手性羟基

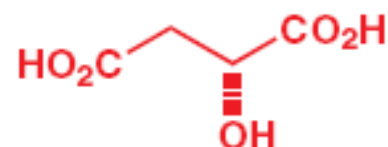
类似的策略可以制备其他的手性醇



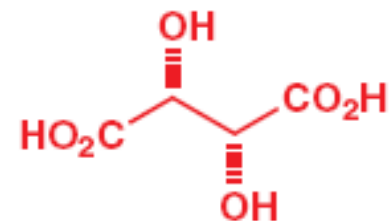
(*S*)-(+)-lactic acid



(*S*)-(+)-mandelic acid



(*R*)-(-)-malic acid

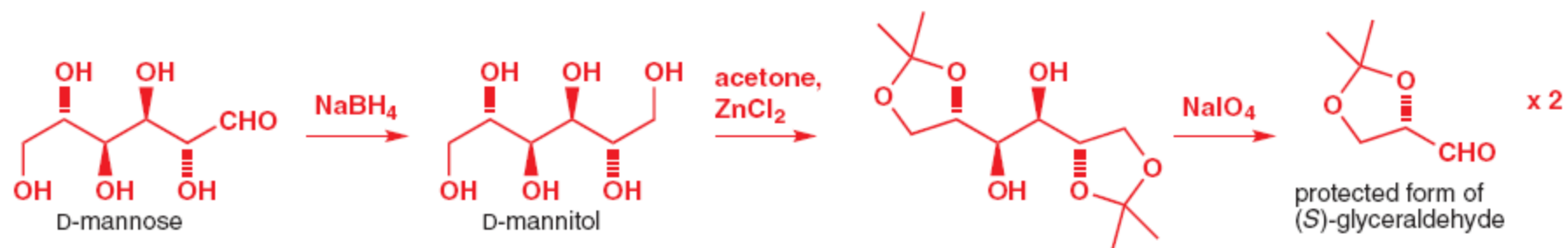


(*R,R*)-(+)-tartaric acid
[= L-tartaric acid]



手性池策略

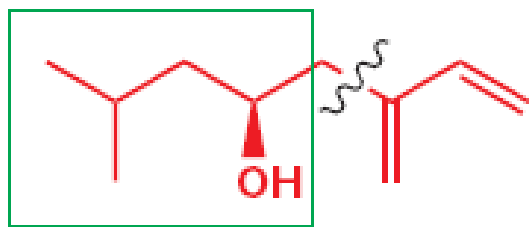
从糖类分子出发，制备手性醇



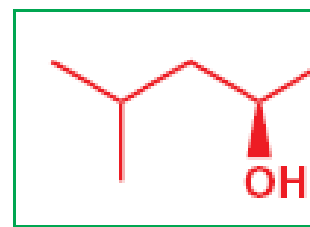


手性池策略

structural similarity with leucine



(S)-(-)-ipsenol



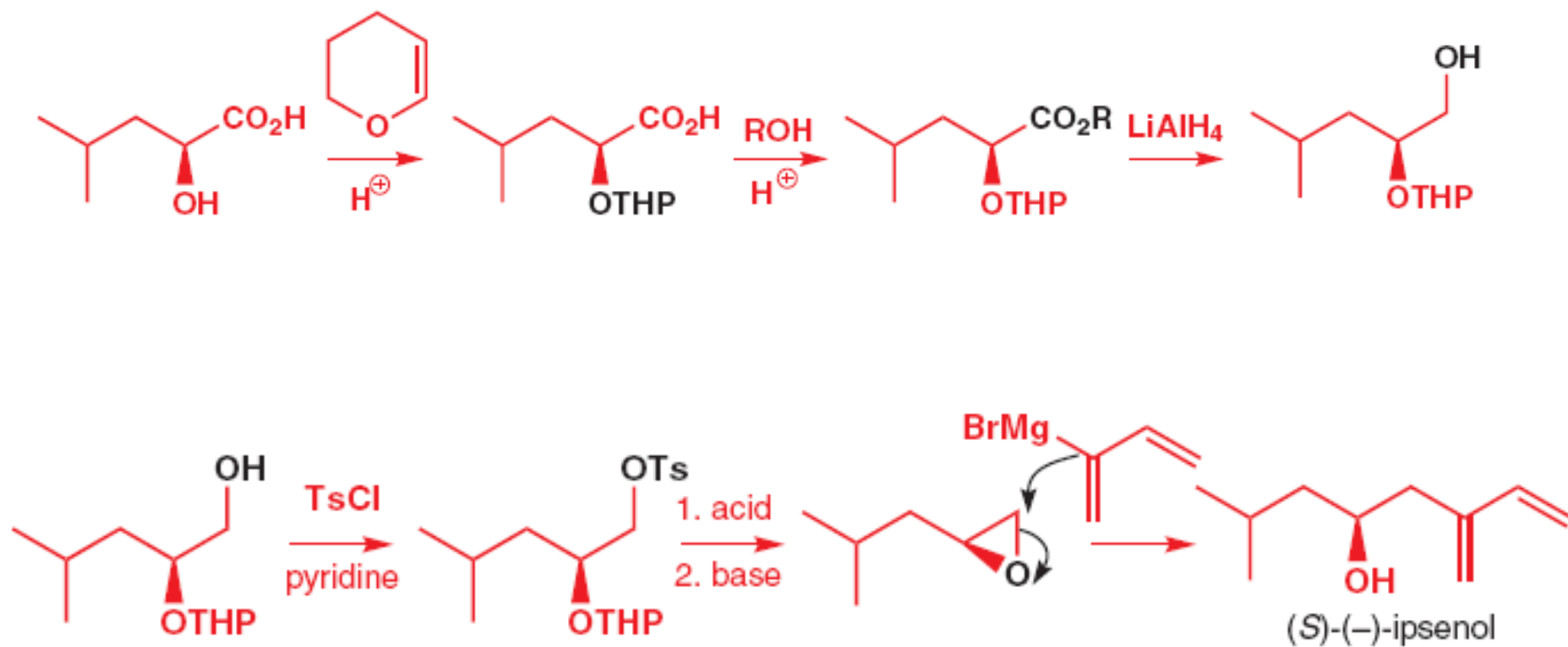
derivative of (S)-leucine

+



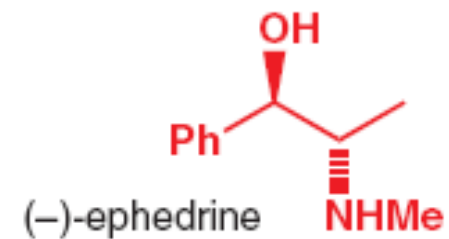
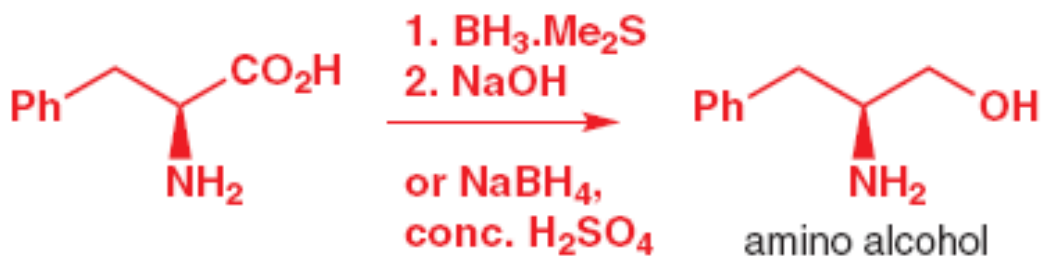


手性池策略





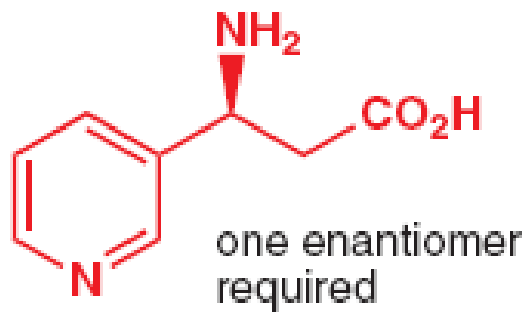
从天然氨基酸合成手性原料



麻黄碱,
手性池



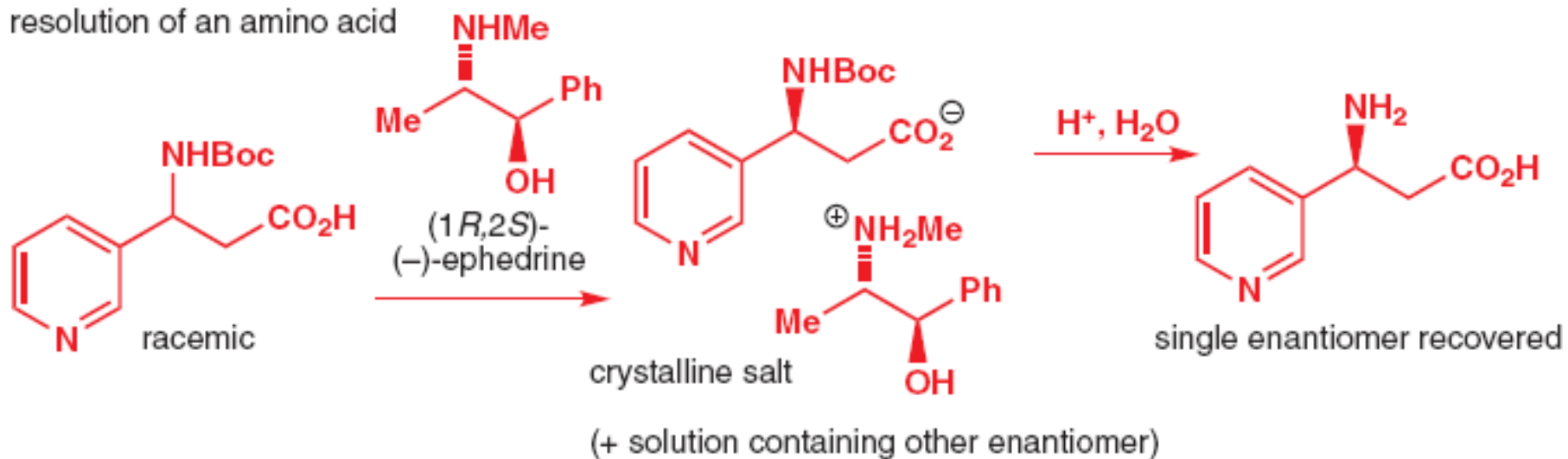
如何不对称合成下面的化合物





手性拆分策略

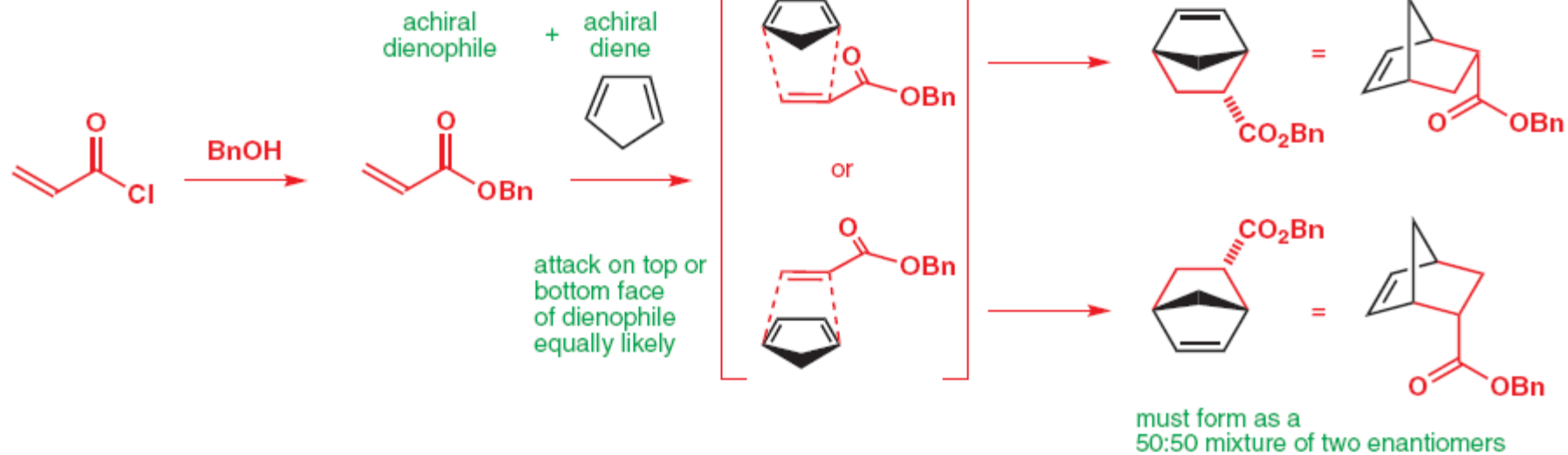
resolution of an amino acid





手性辅基策略

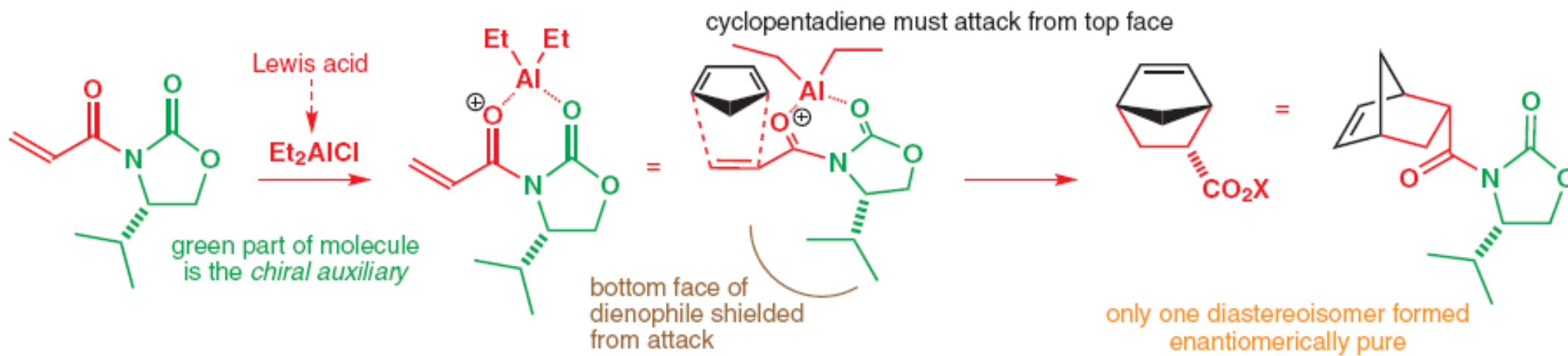
Diels–Alder reaction gives a racemic product





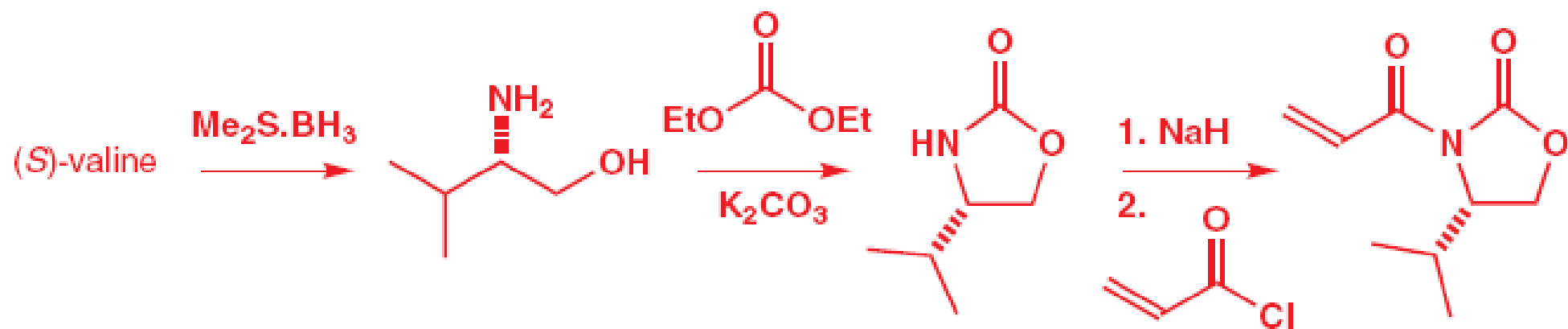
手性辅基策略

手性辅基控制的不对称Diels-Alder反应



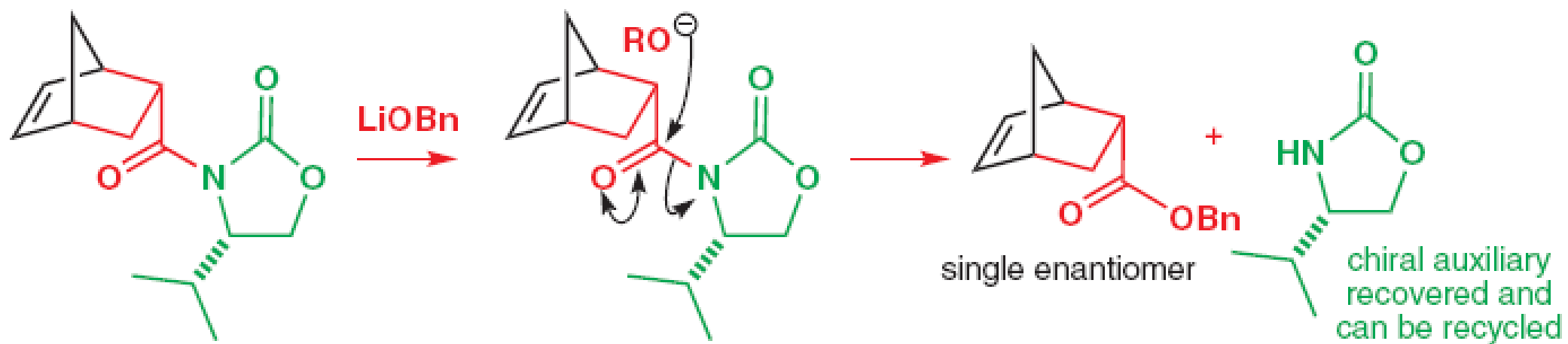


手性辅基的合成





手性辅基控制的不对称Diels-Alder反应

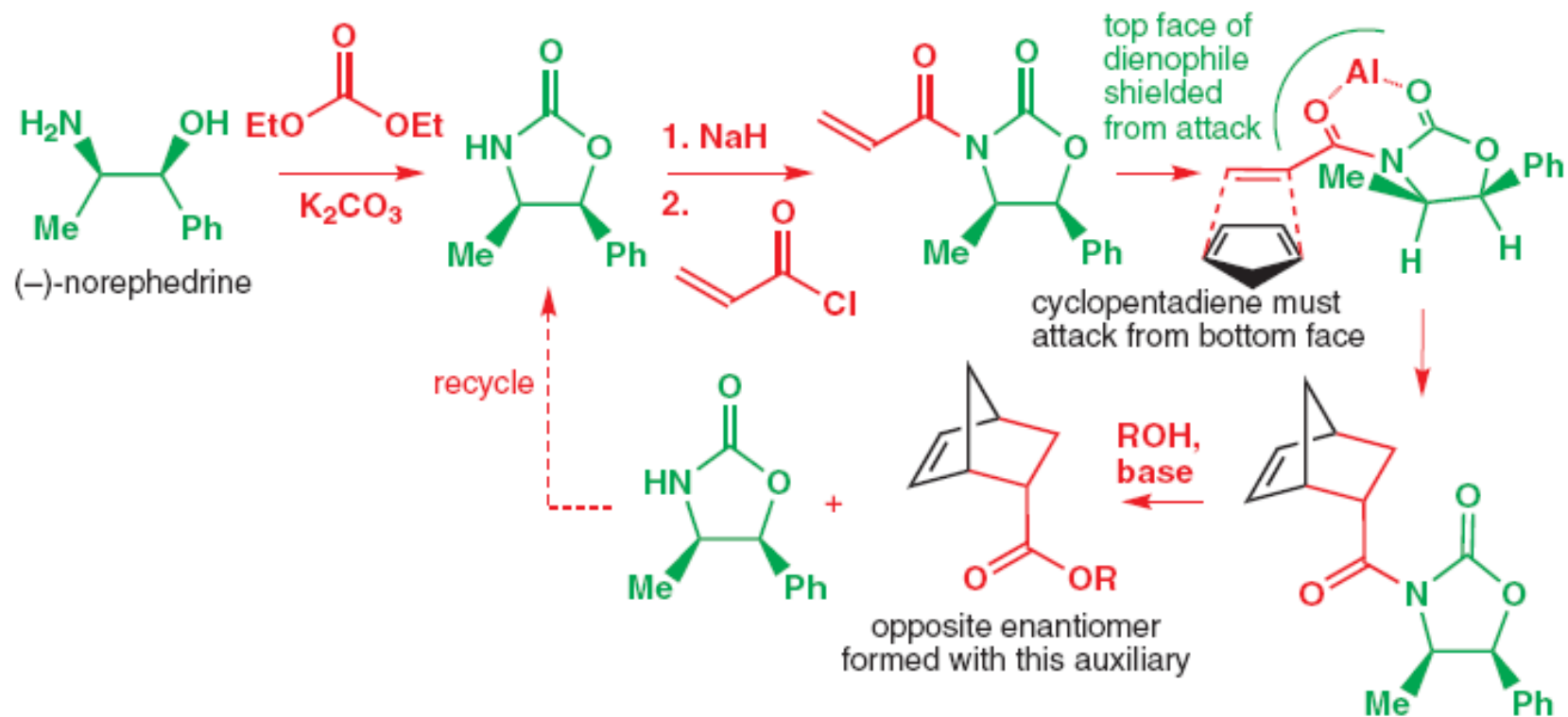


辅基回收



手性辅基策略

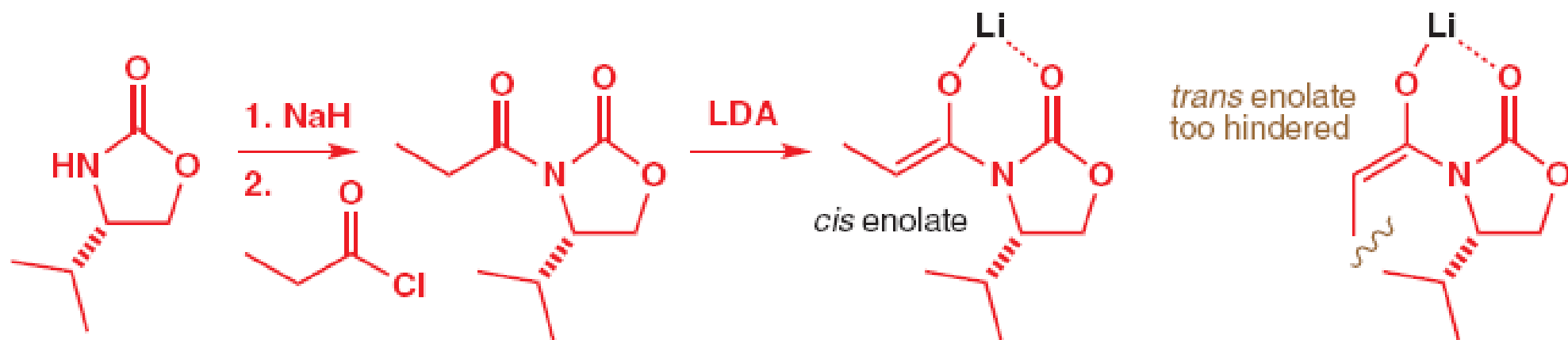
对映体的手性辅基策略





手性辅基策略

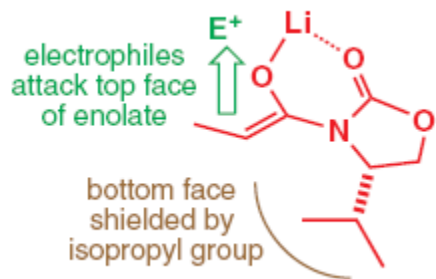
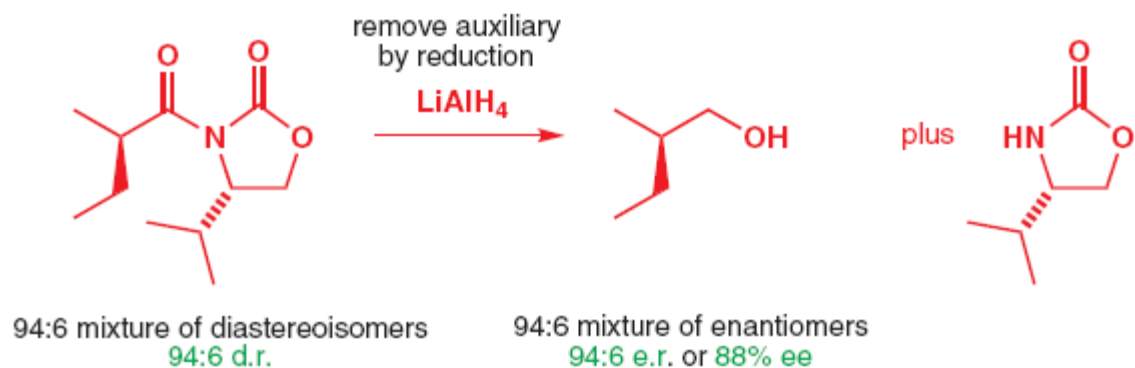
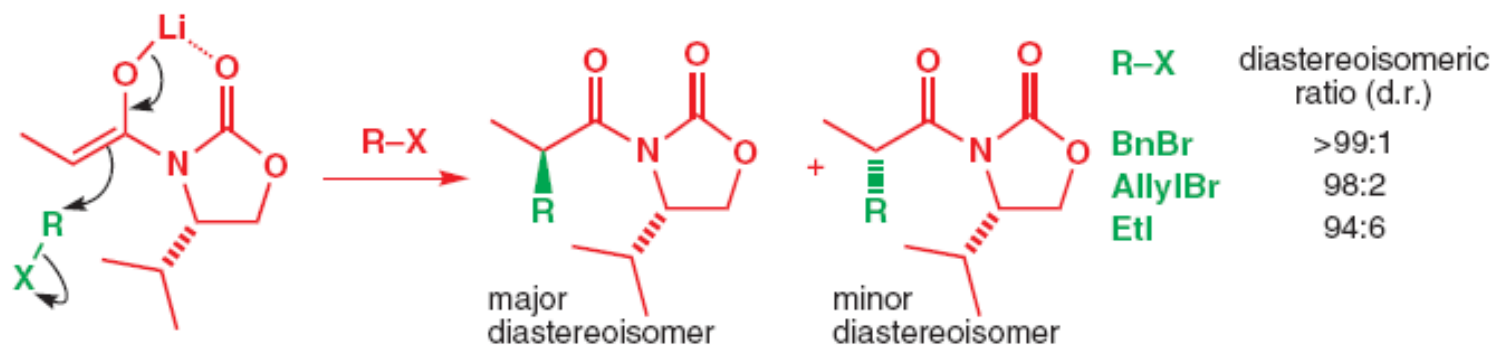
手性辅基控制的aldol反应





手性辅基策略

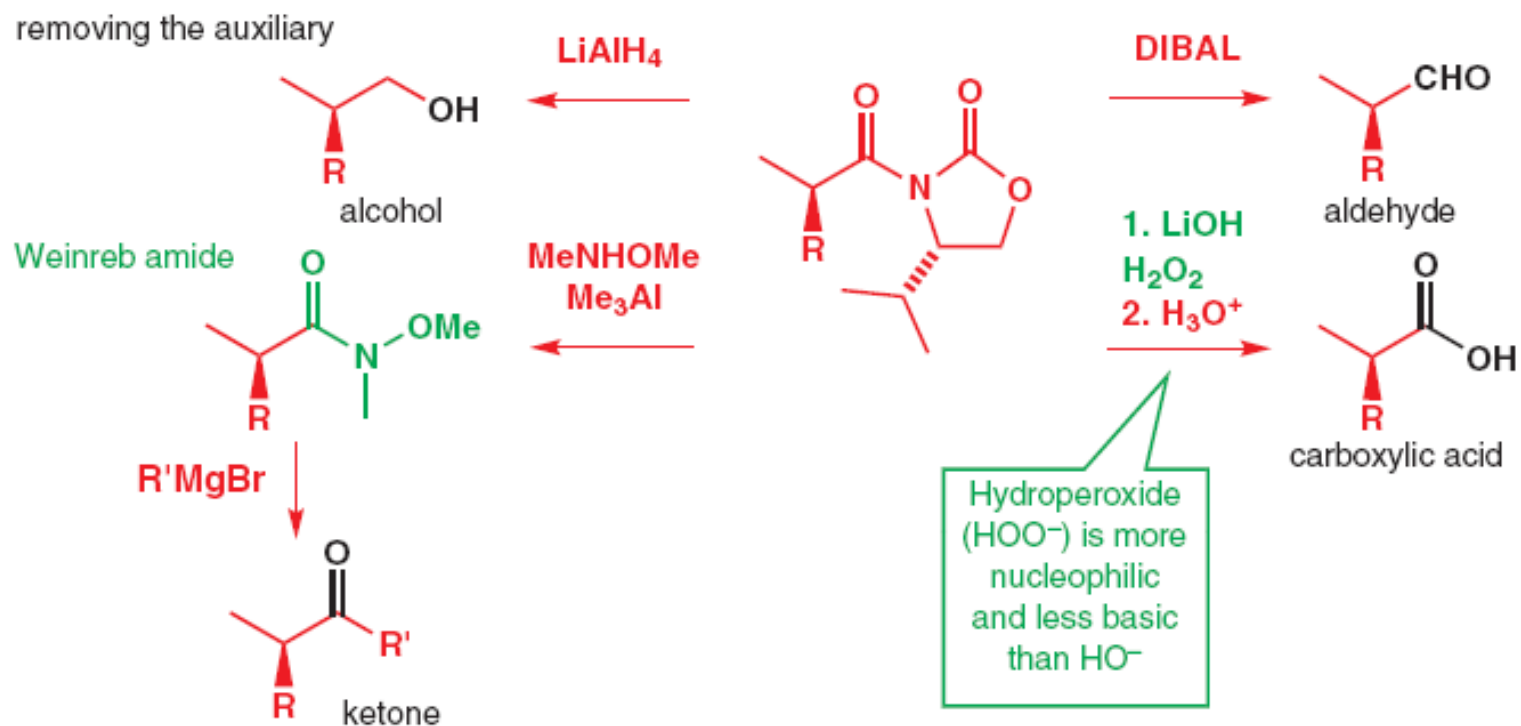
手性辅基控制的aldol反应





手性辅基策略

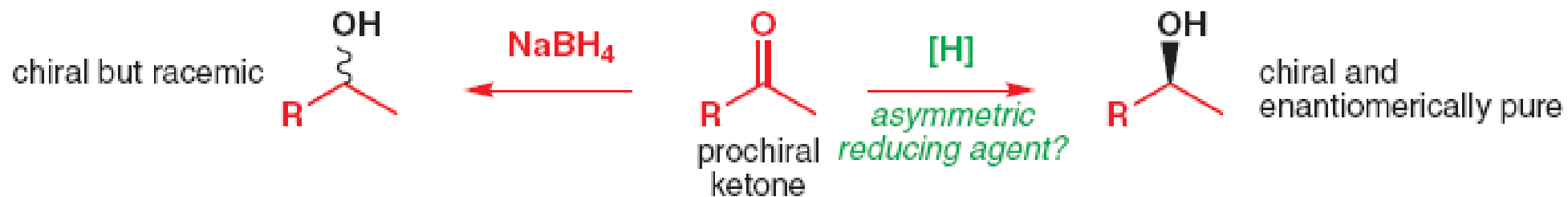
手性辅基控制的aldol反应





不对称催化策略

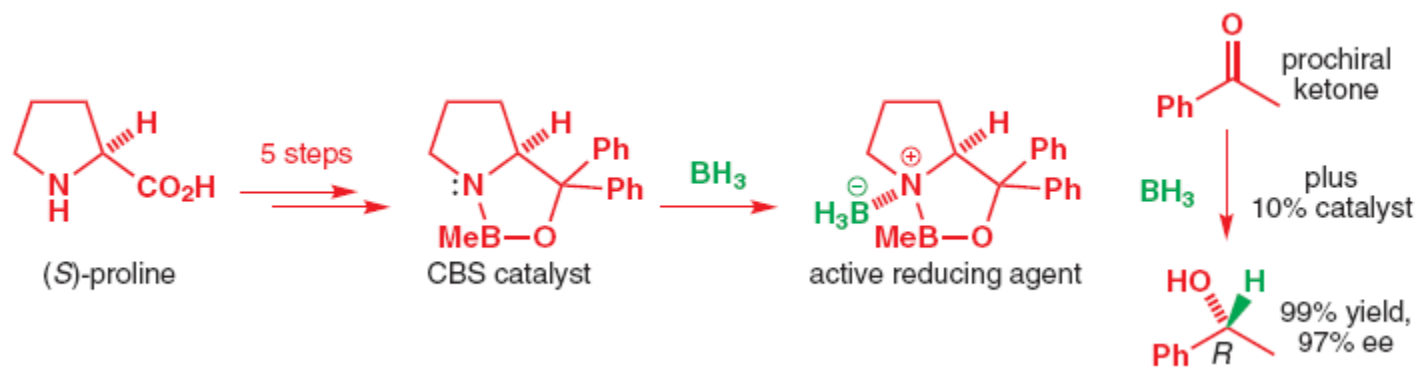
手性配体-金属复合物不对称催化实现不对称合成





不对称催化策略

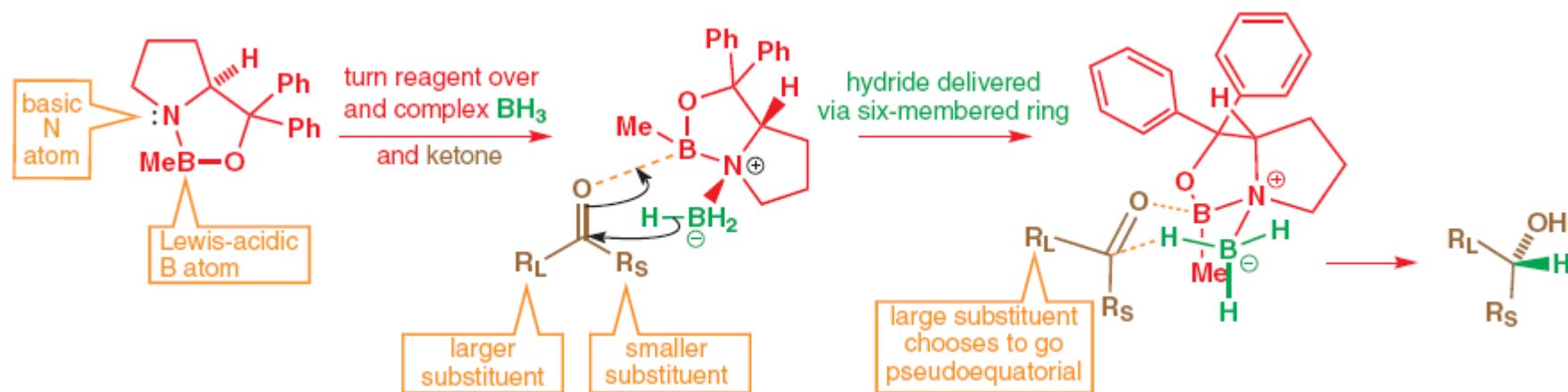
CBS催化不对称氢化





不对称催化策略

CBS催化不对称氢化机理

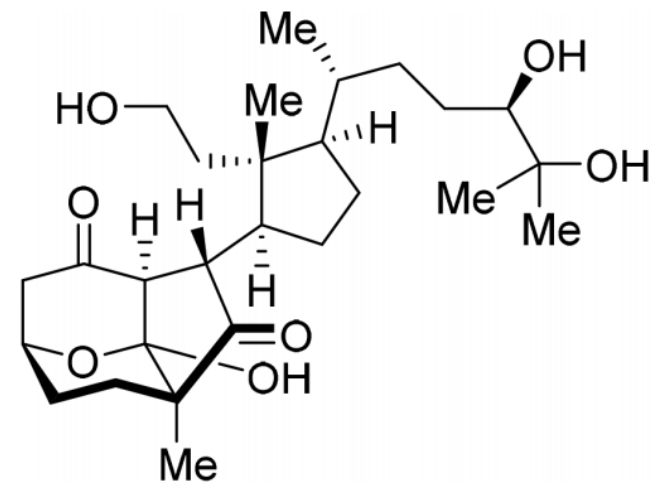
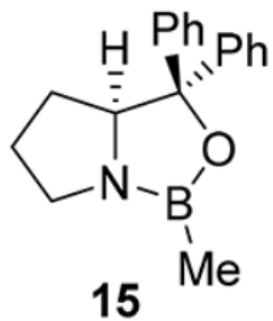
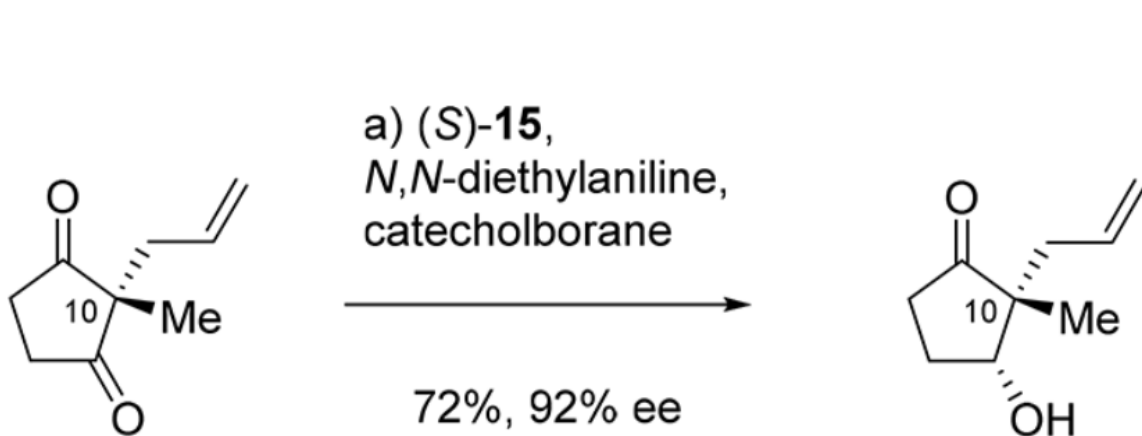


去对称化的一种方法



不对称催化策略

CBS催化不对称还原，去对称化的一种方法

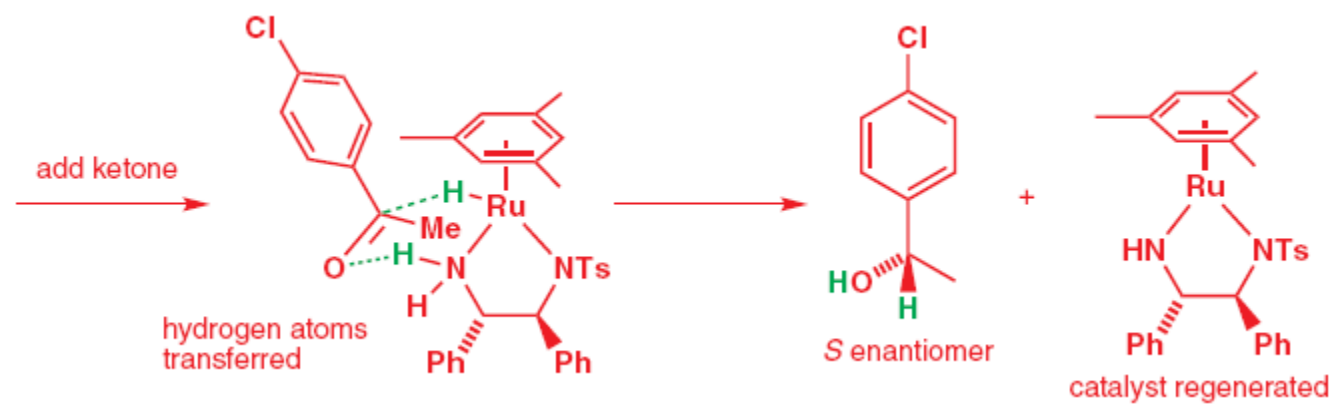


1: aplysiasecosterol A



不对称催化策略

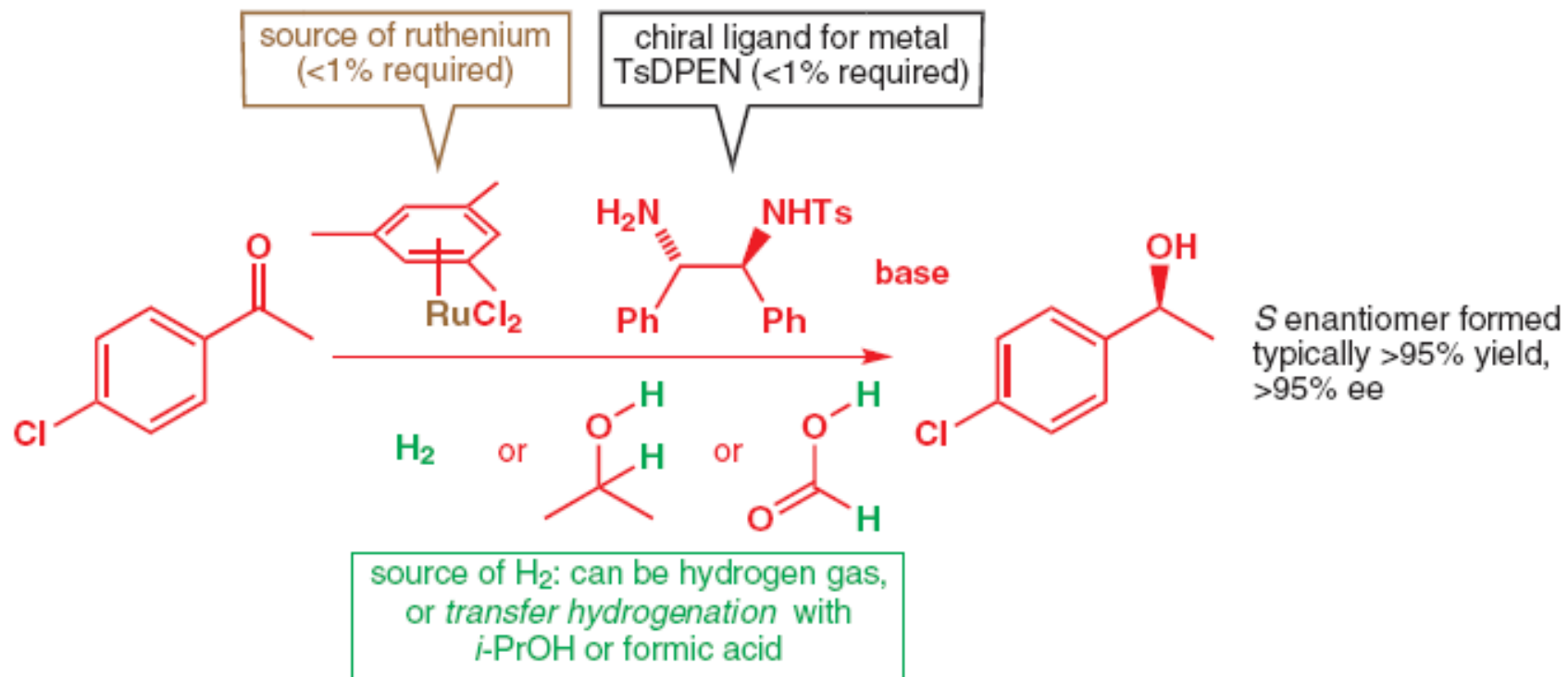
金属不对称催化





不对称催化策略

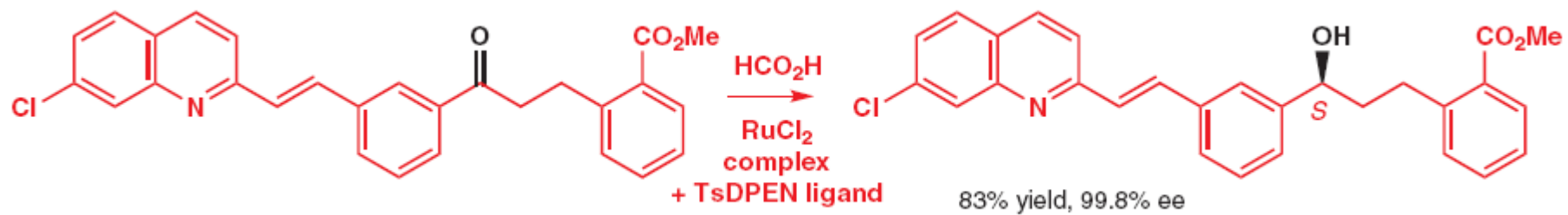
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不对称催化策略

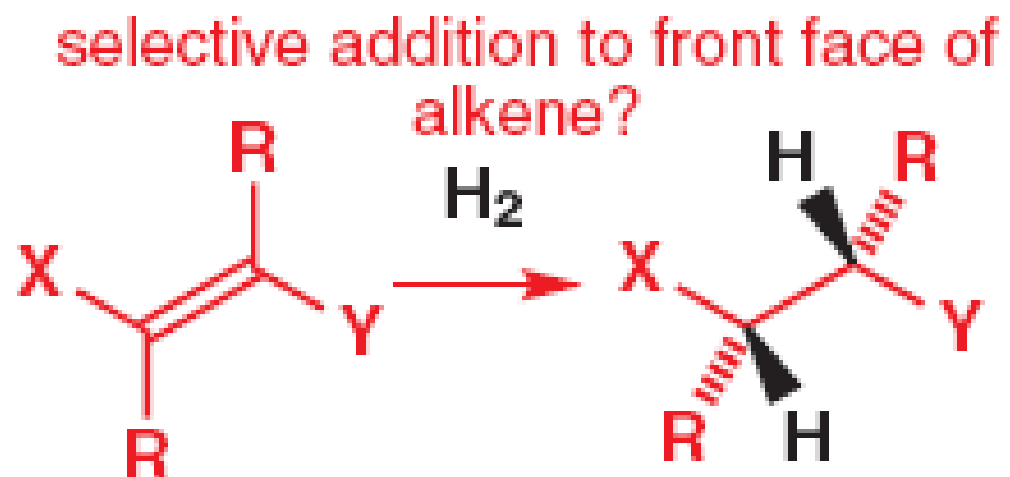
金属不对称催化





不对称催化策略

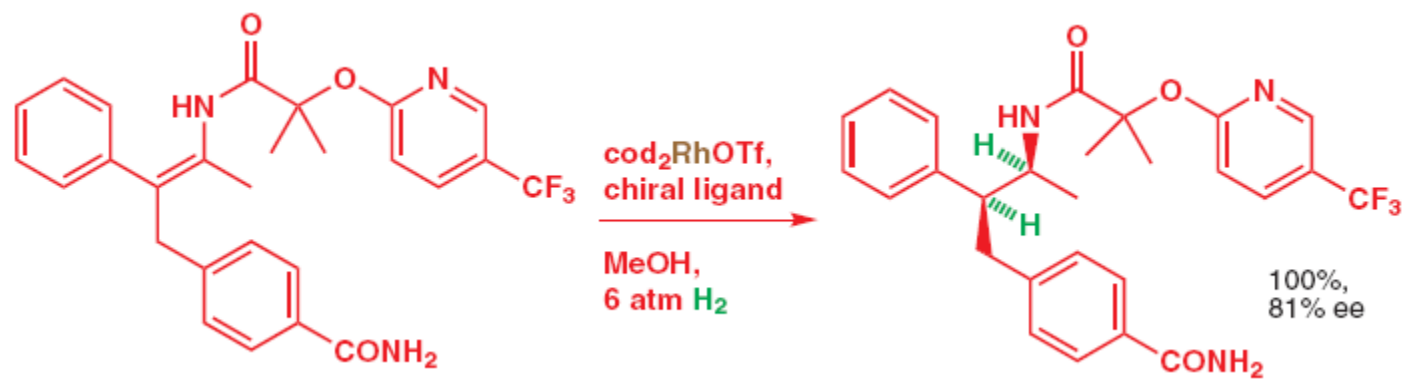
不对称氢化碳碳双键





不对称催化策略

不对称氢化烯烃

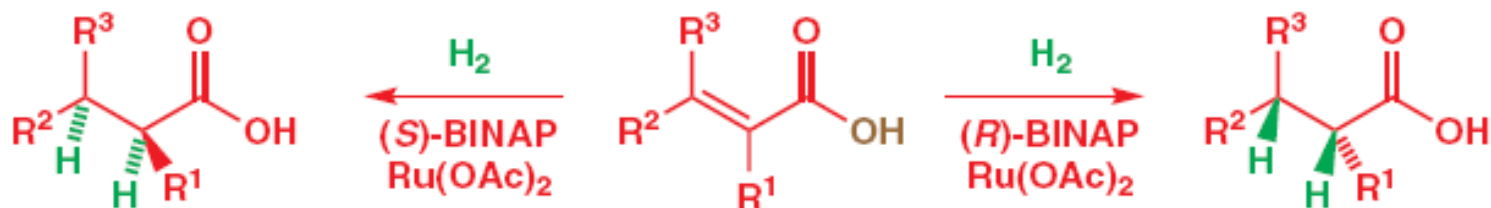




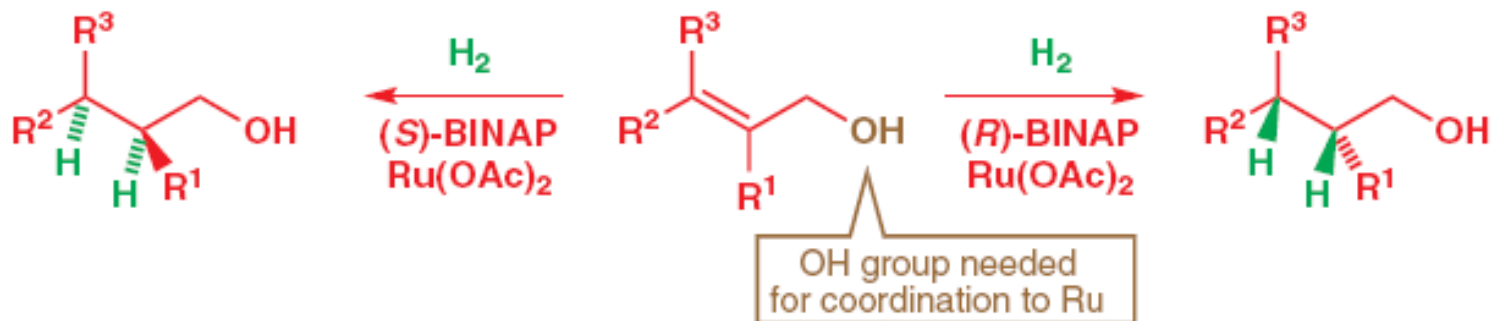
不对称催化策略

不对称氢化

Ru-catalysed asymmetric hydrogenation of unsaturated carboxylic acids



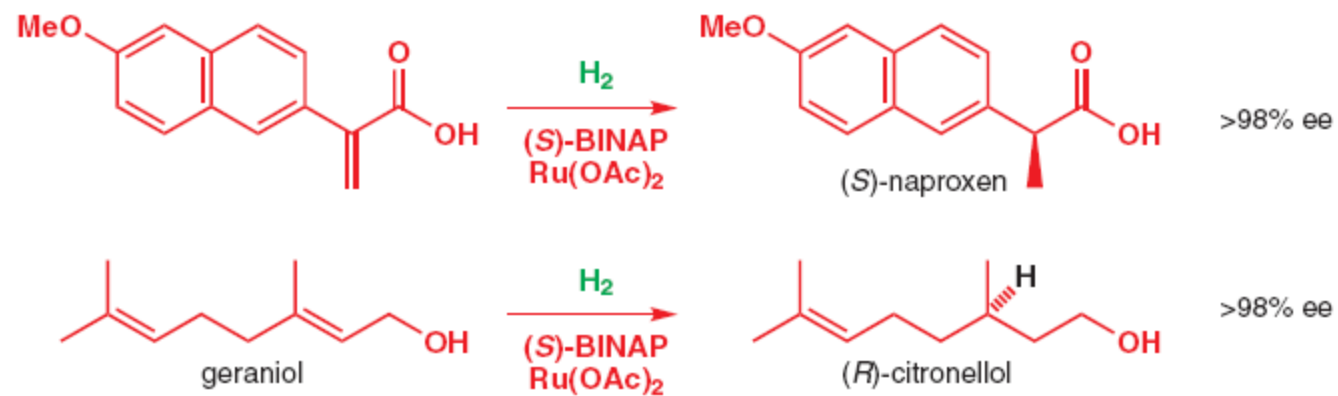
Ru-catalysed asymmetric hydrogenation of allylic alcohols





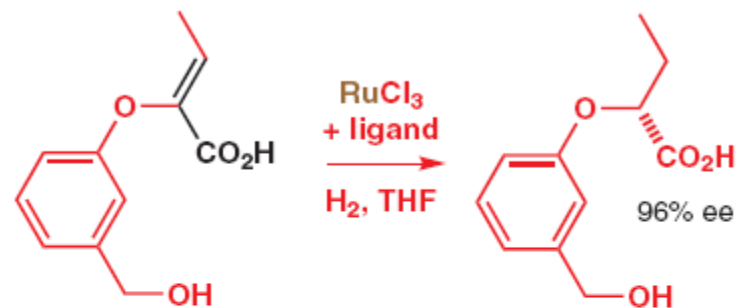
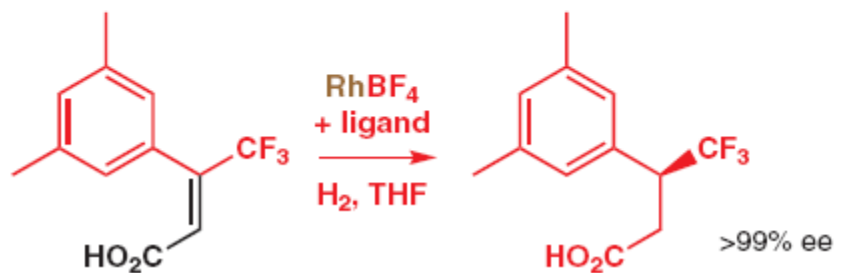
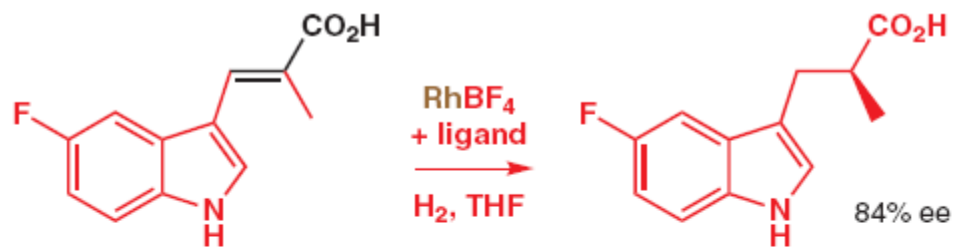
不对称催化策略

不对称氢化烯烃





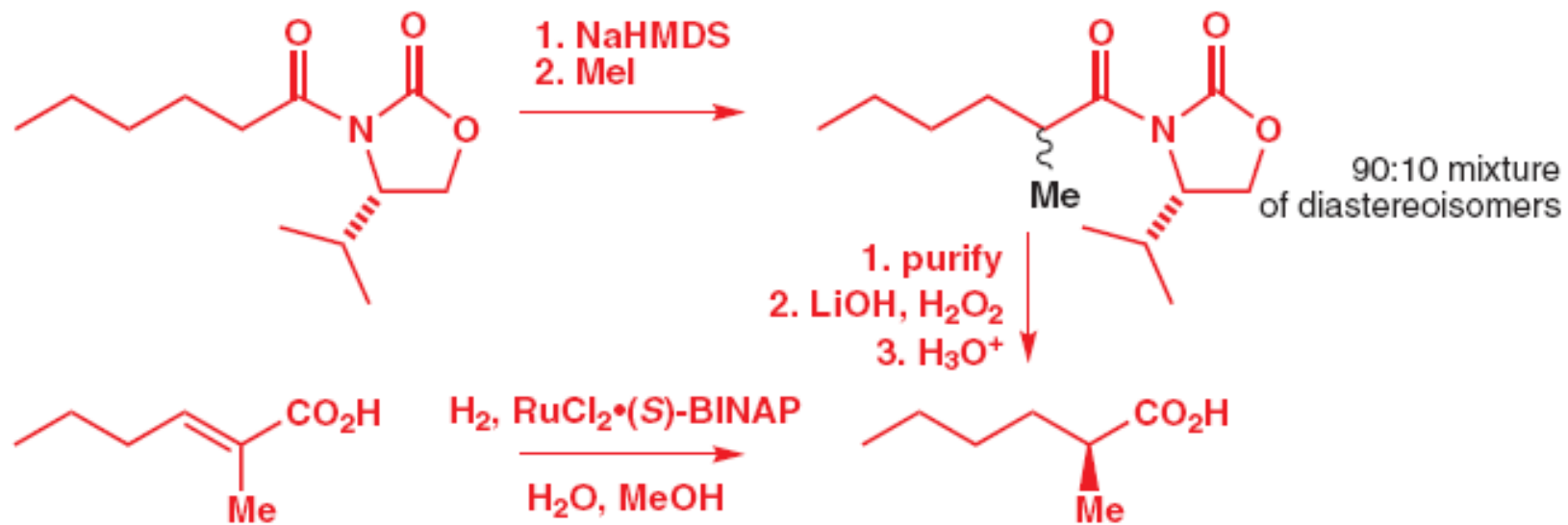
不对称催化策略





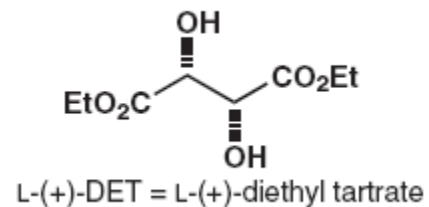
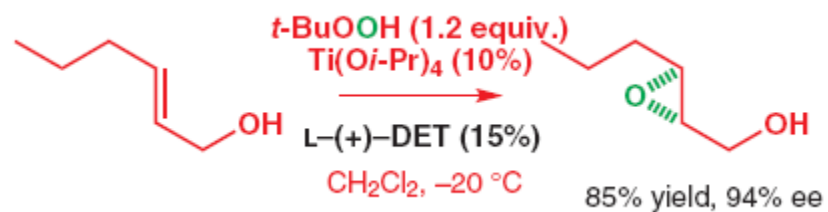
不对称催化策略

不对称氢化vs手性辅基





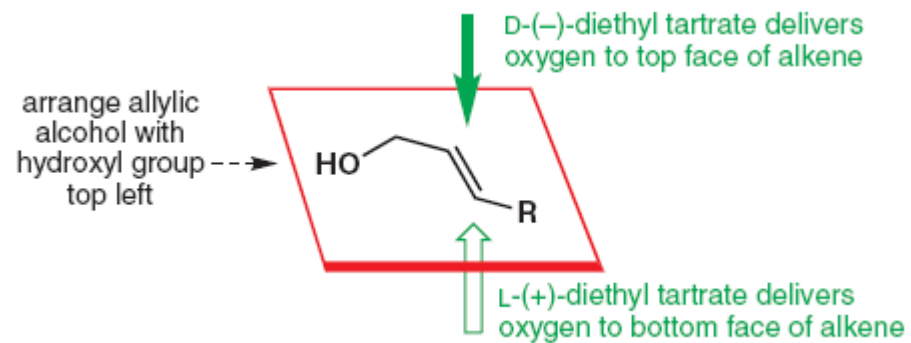
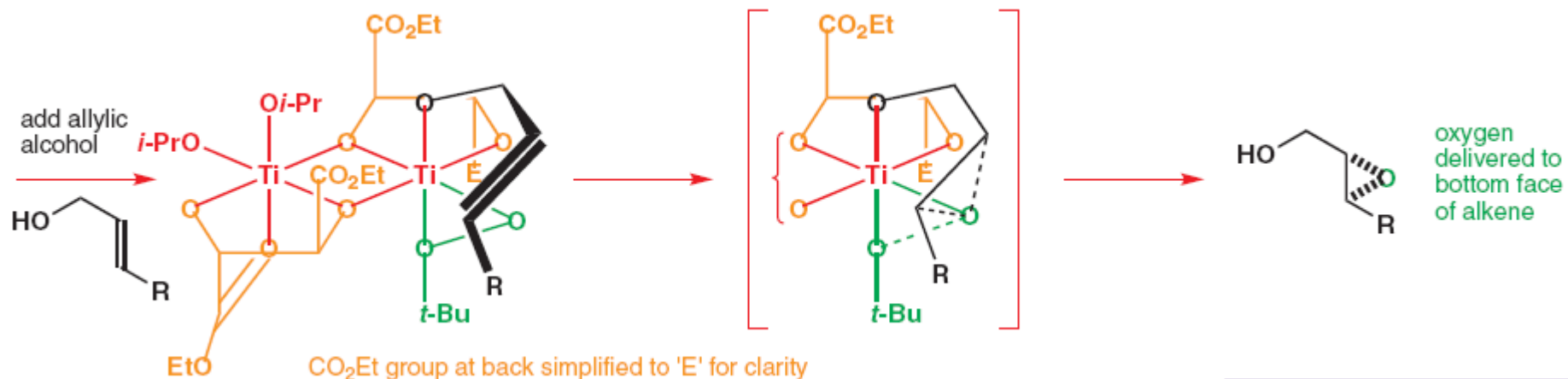
不对称环氧化





不对称催化策略

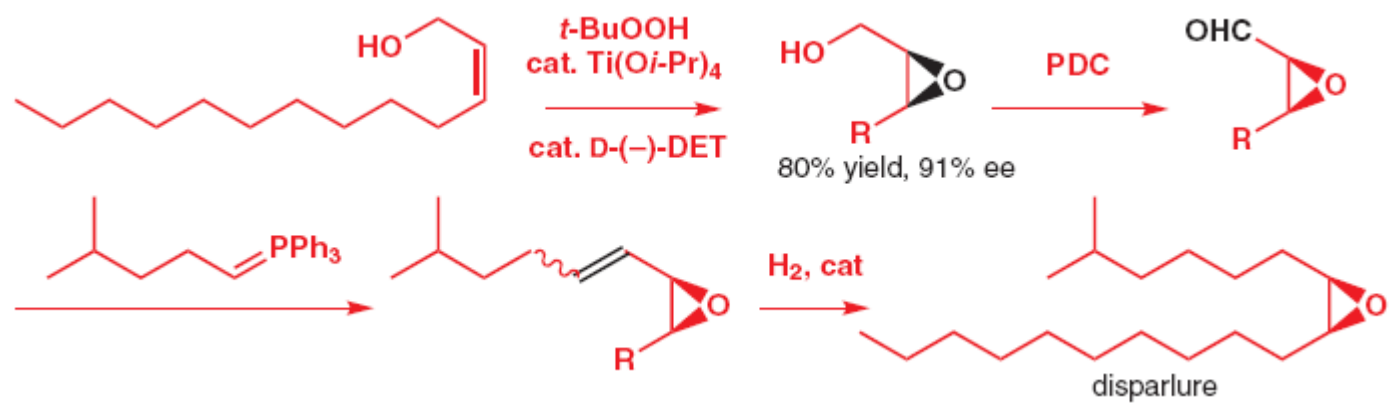
不对称环氧化机理





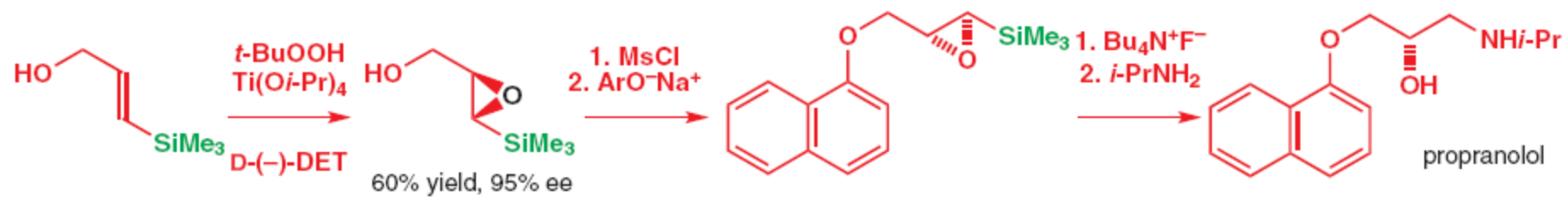
不对称催化策略

不对称环氧化应用



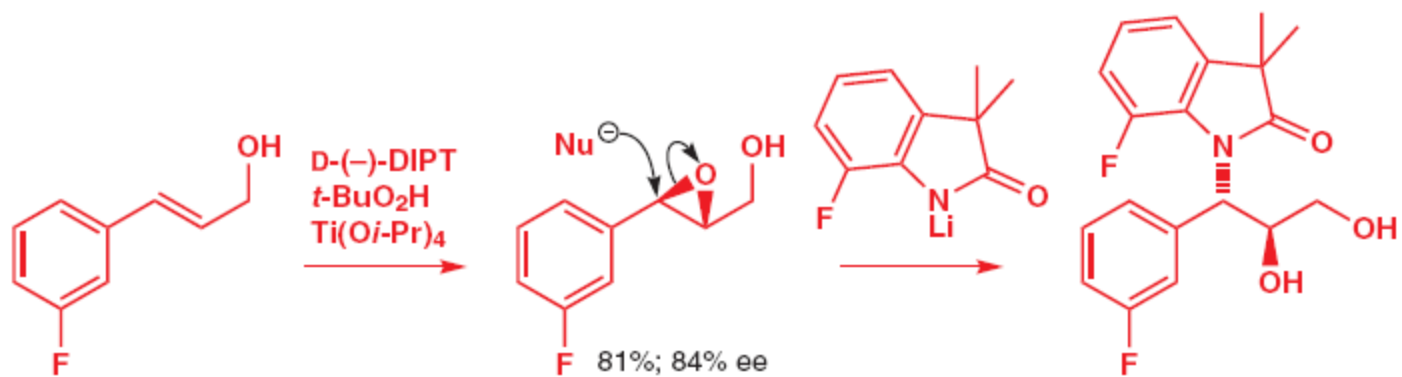


不对称催化策略





不对称催化策略

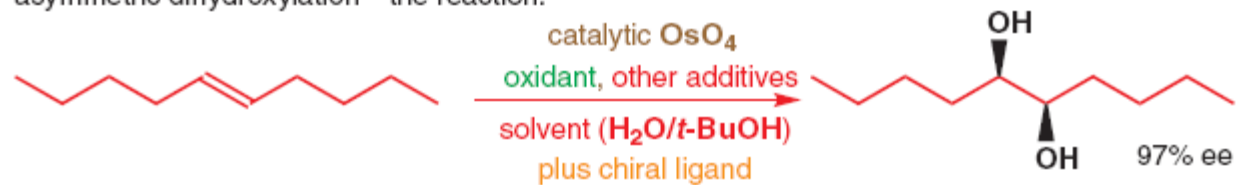




不对称催化策略

不对称双羟化

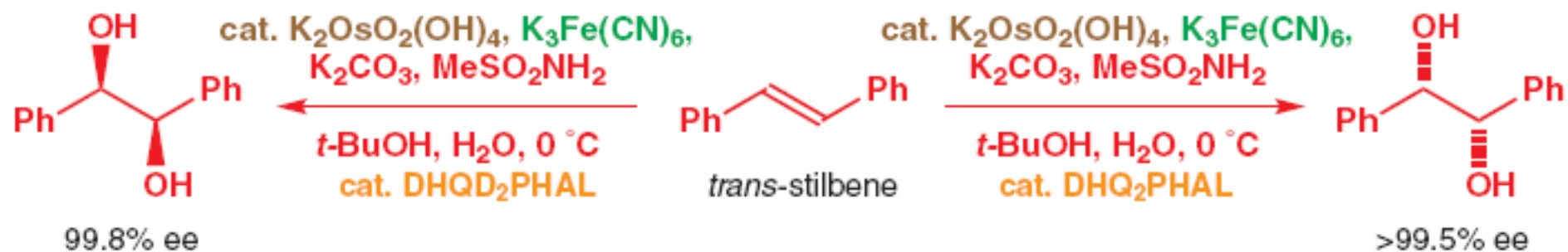
asymmetric dihydroxylation—the reaction:





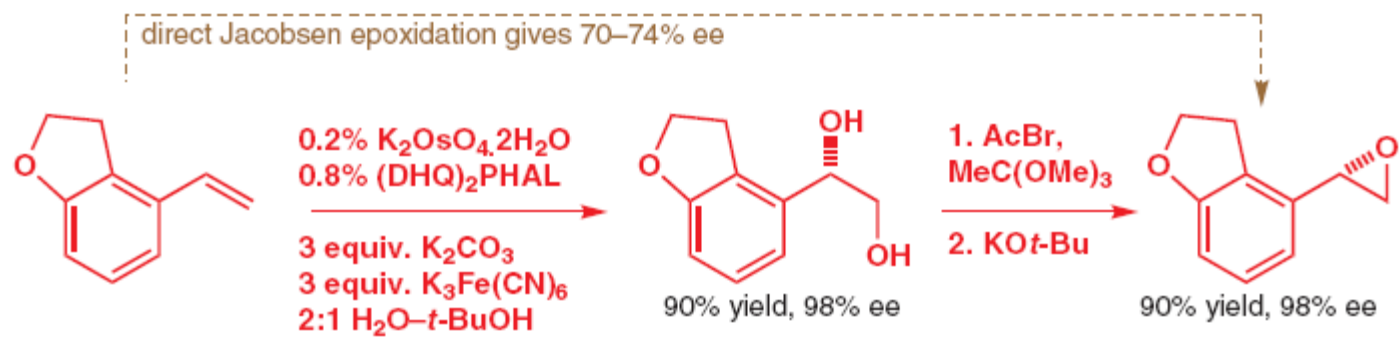
不对称催化策略

不对称双羟化



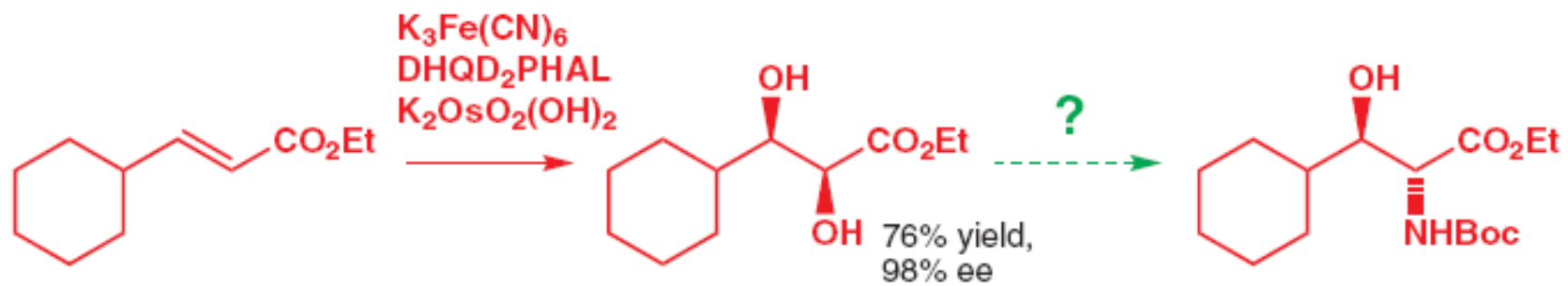


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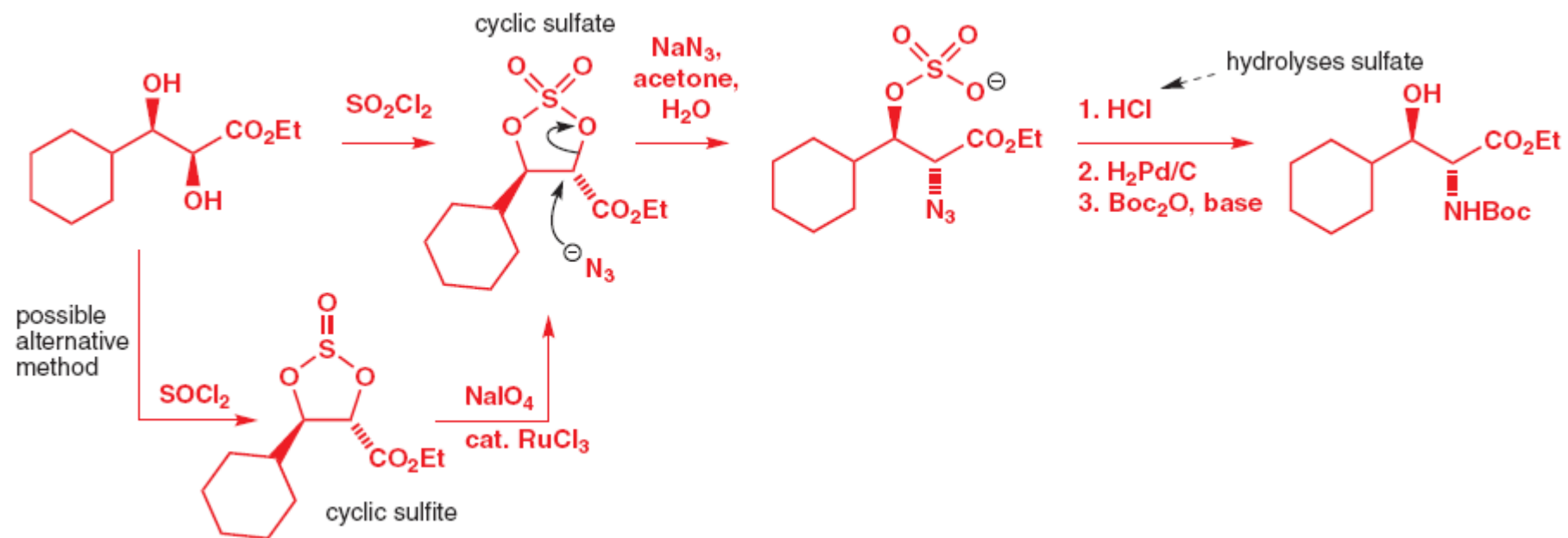


不对称催化策略





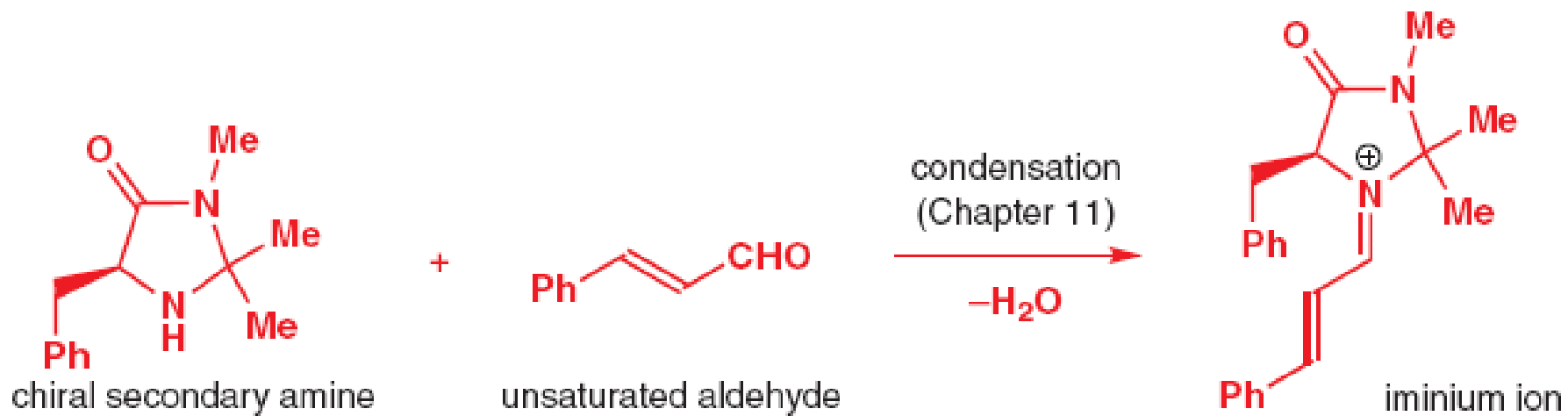
不对称催化策略





不对称催化策略

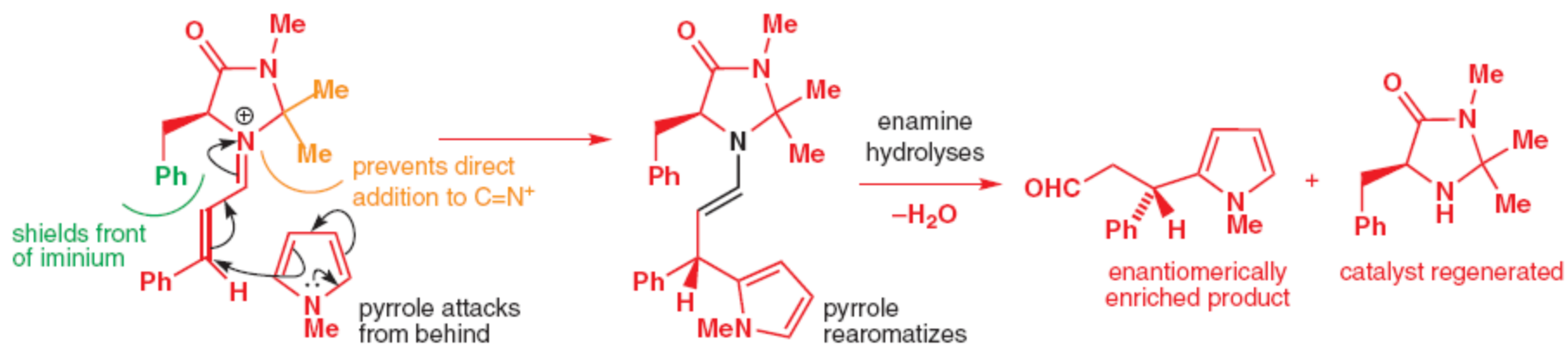
不对称小分子催化





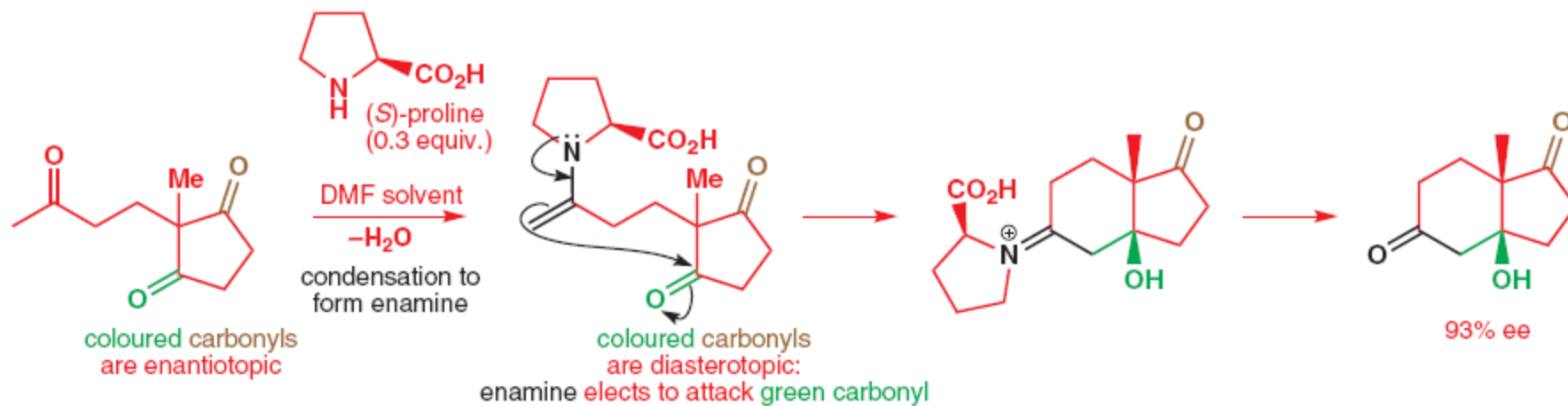
不对称催化策略

不对称有机小分子催化





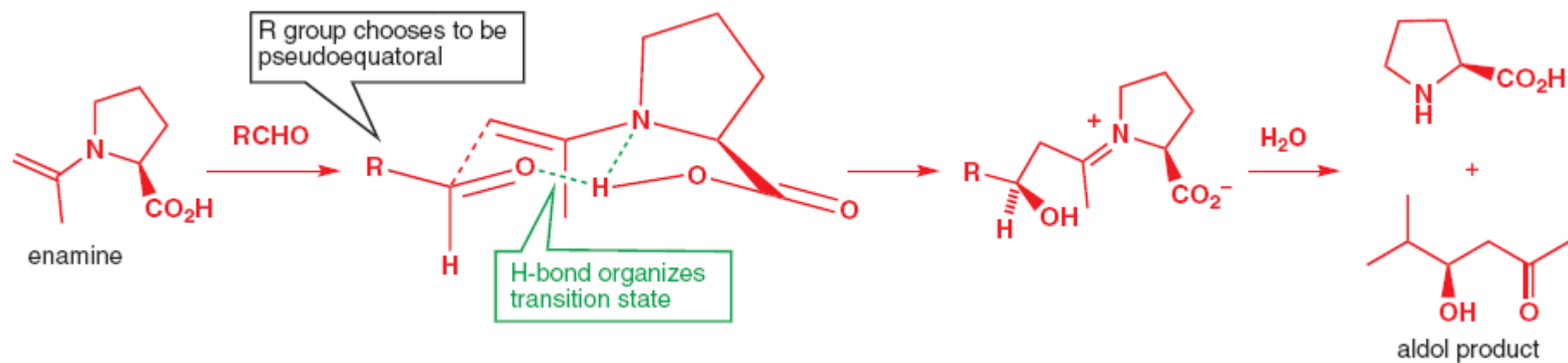
不对称催化策略





不对称催化策略

脯氨酸小分子催化不对称aldol反应的机理





不对称催化策略

脯氨酸小分子催化不对称aldol反应的机理

